

Clinical Risk Assessment Report: Patient 26

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.49x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.5%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.08 [Top 20%] (-1.5%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-1.1%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.51 [Avg (32th %ile)] (-1.0%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.28 [Avg (59th %ile)] (-0.8%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.01 [Top 20%] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.9 [Avg (74th %ile)] (-0.6%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.45 [Bottom 22%] (-0.6%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -0.58 [Bottom 23%] (-0.5%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)

